



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Subject: Advanced Digital Signal Processing

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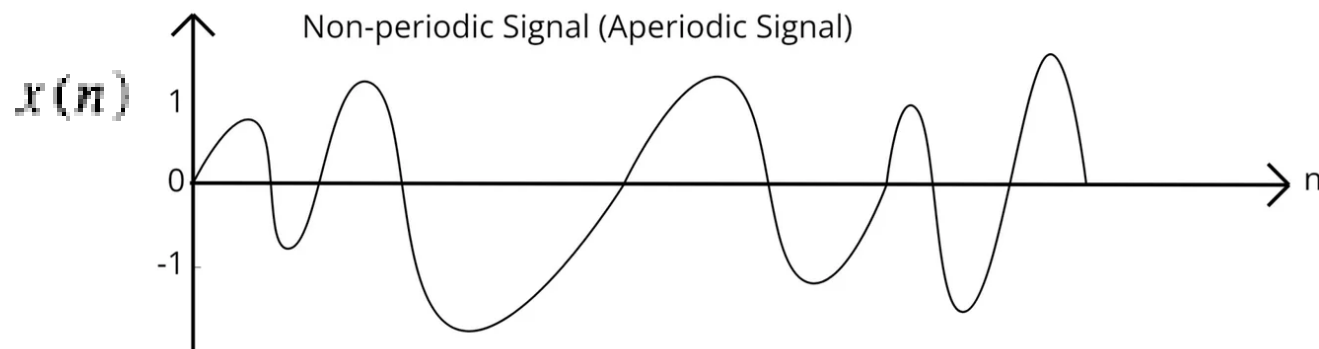
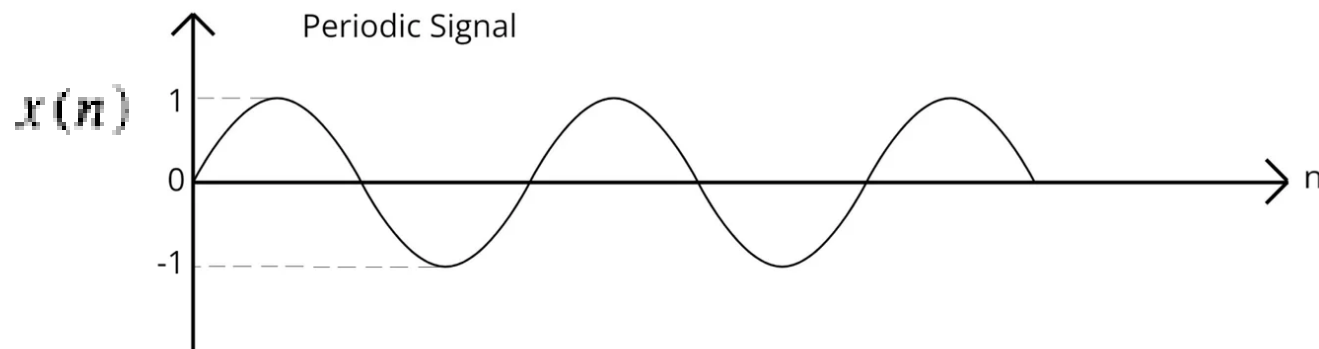
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Periodic signals and aperiodic signals.

signal $x(n)$ is periodic with period N ($N > 0$) if and only if

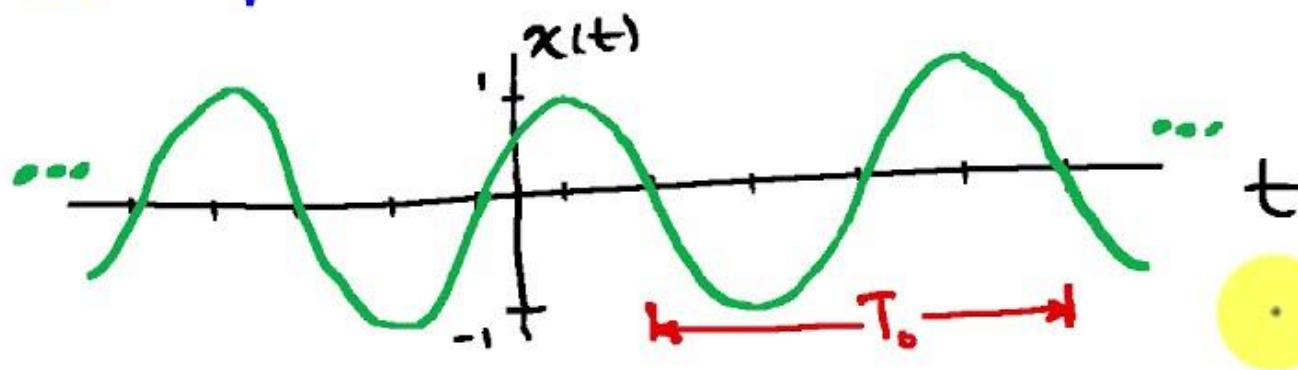
$$x(n + N) = x(n) \text{ for all } n \quad (2.1.20)$$

The smallest value of N for which (2.1.20) holds is called the (fundamental) period. If there is no value of N that satisfies (2.1.20), the signal is called *nonperiodic* or *aperiodic*.



Example: $x(t) = \cos(2\pi f_0 t + \phi)$

3



f_0 Hz $\Rightarrow f_0$ cycles in 1 sec
 $\Rightarrow$ each cycle $T_0 = \frac{1}{f_0}$ sec

$$\begin{aligned} \text{Math: } x(t+T_0) &= \cos(2\pi f_0(t+T_0) + \phi) \\ &= \cos(2\pi f_0 t + \underline{2\pi f_0 T_0} + \phi) \\ &= \cos(2\pi f_0 t + \phi) \quad \text{for } T_0 = \frac{1}{f_0} \\ &= x(t) \quad \text{for all } t \end{aligned}$$

Fundamental period: smallest T_0

Periodic signals and aperiodic signals.

the sinusoidal signal of the form

$$x(n) = A \sin 2\pi f_0 n \quad (2.1.21)$$

is periodic when f_0 is a rational number, that is, if f_0 can be expressed as

$$f_0 = \frac{k}{N} \quad (2.1.22)$$

where k and N are integers.

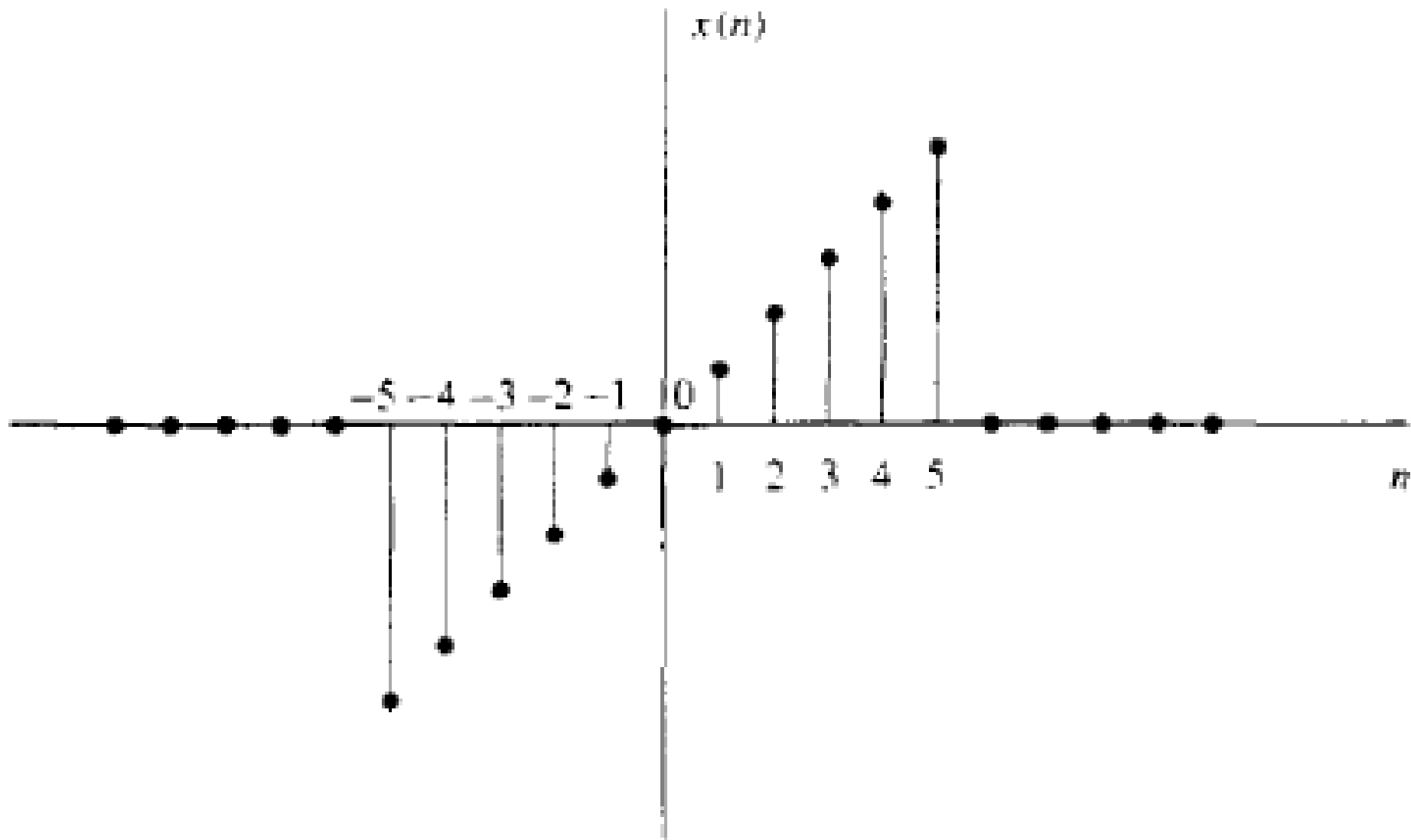
Symmetric (even) and antisymmetric (odd) signals. A real-valued signal $x(n)$ is called symmetric (even) if

$$x(-n) = x(n) \quad (2.1.24)$$

On the other hand, a signal $x(n)$ is called antisymmetric (odd) if

$$x(-n) = -x(n) \quad (2.1.25)$$

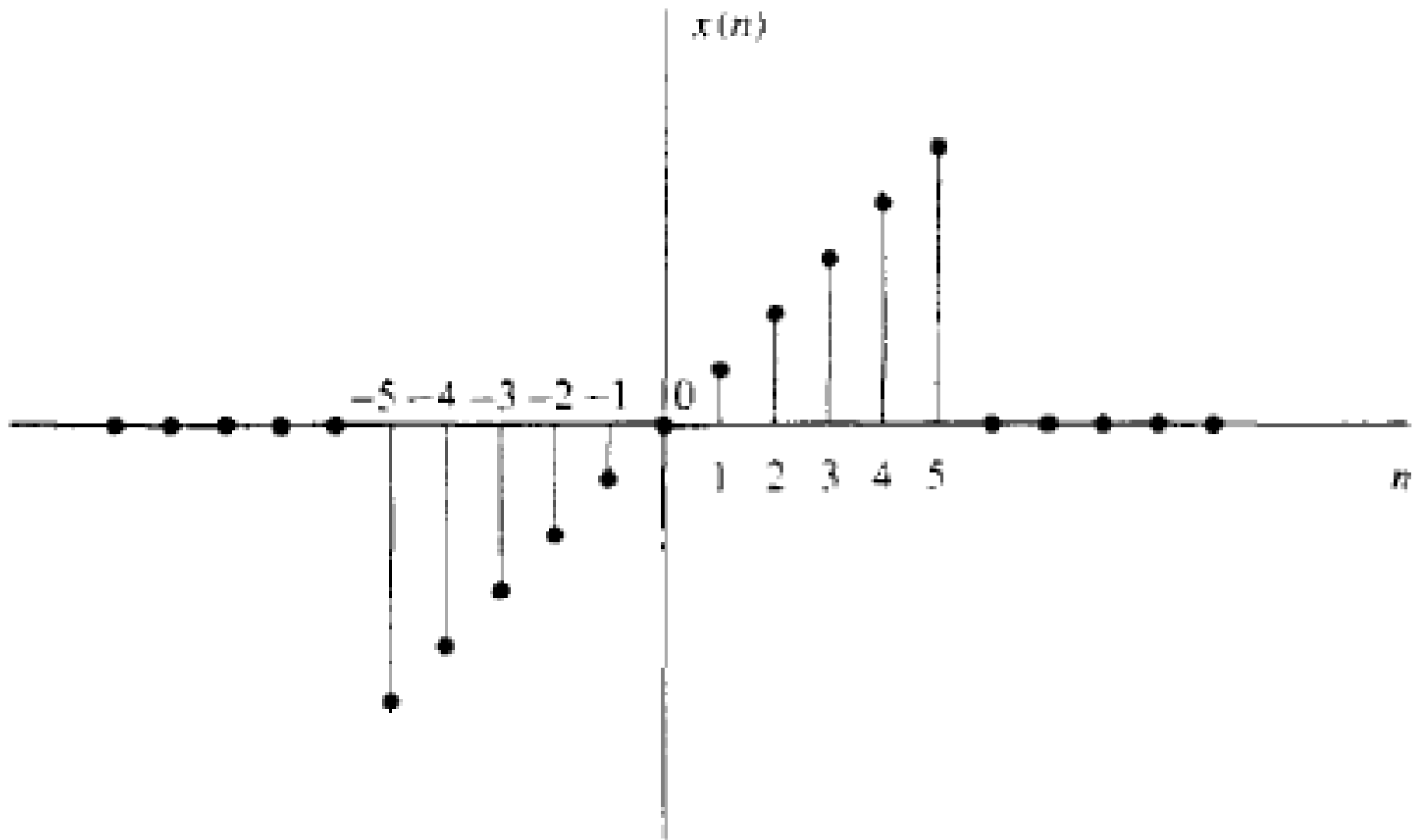
We note that if $x(n)$ is odd, then $x(0) = 0$. Examples of signals with even and odd symmetry are illustrated in Fig. 2.8.



$$x(-n) = -x(n)$$

(b)

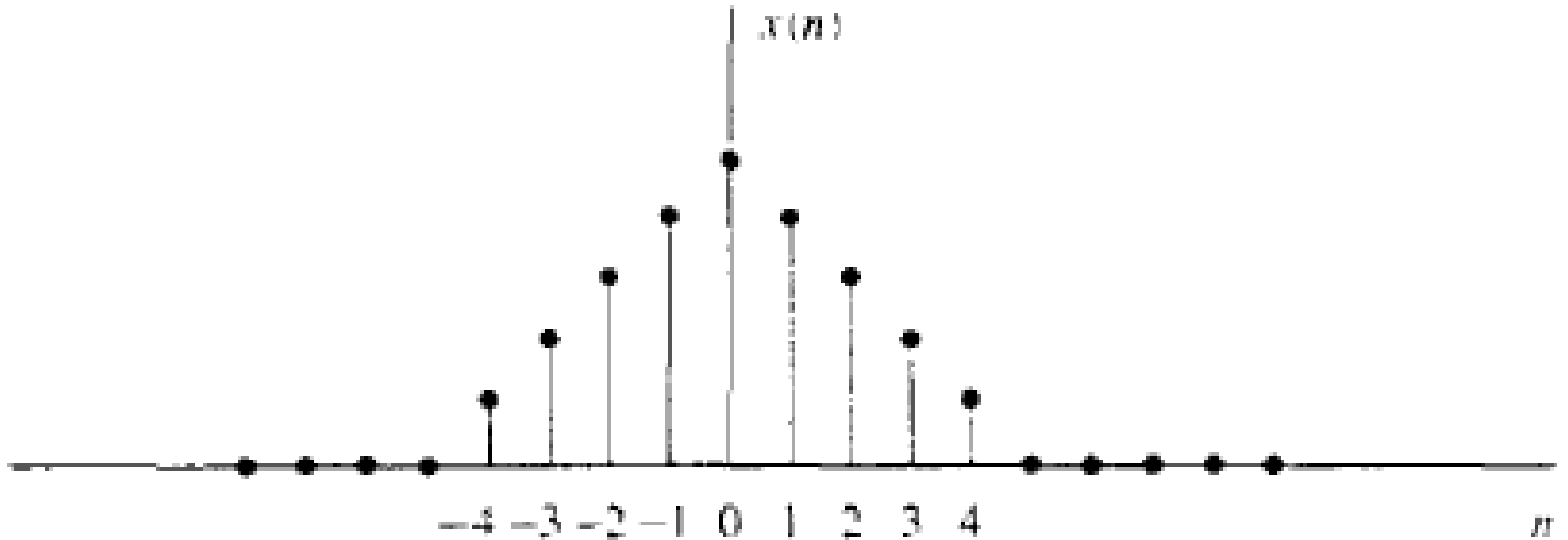
Is this the even signal or odd signal?



$$x(-n) = -x(n)$$

(b)

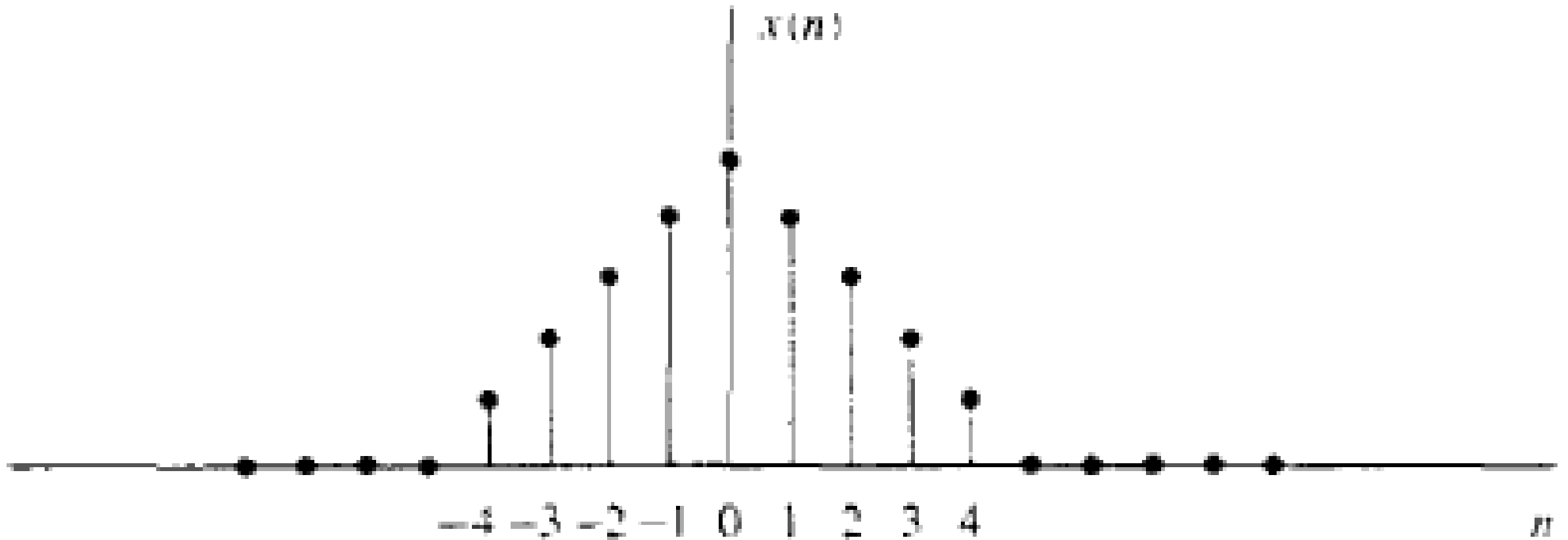
This is the odd signal.



(a)

$$x(-n) = x(n)$$

Is this the even signal or odd signal?



(a)

$$x(-n) = x(n)$$

This is the even signal.

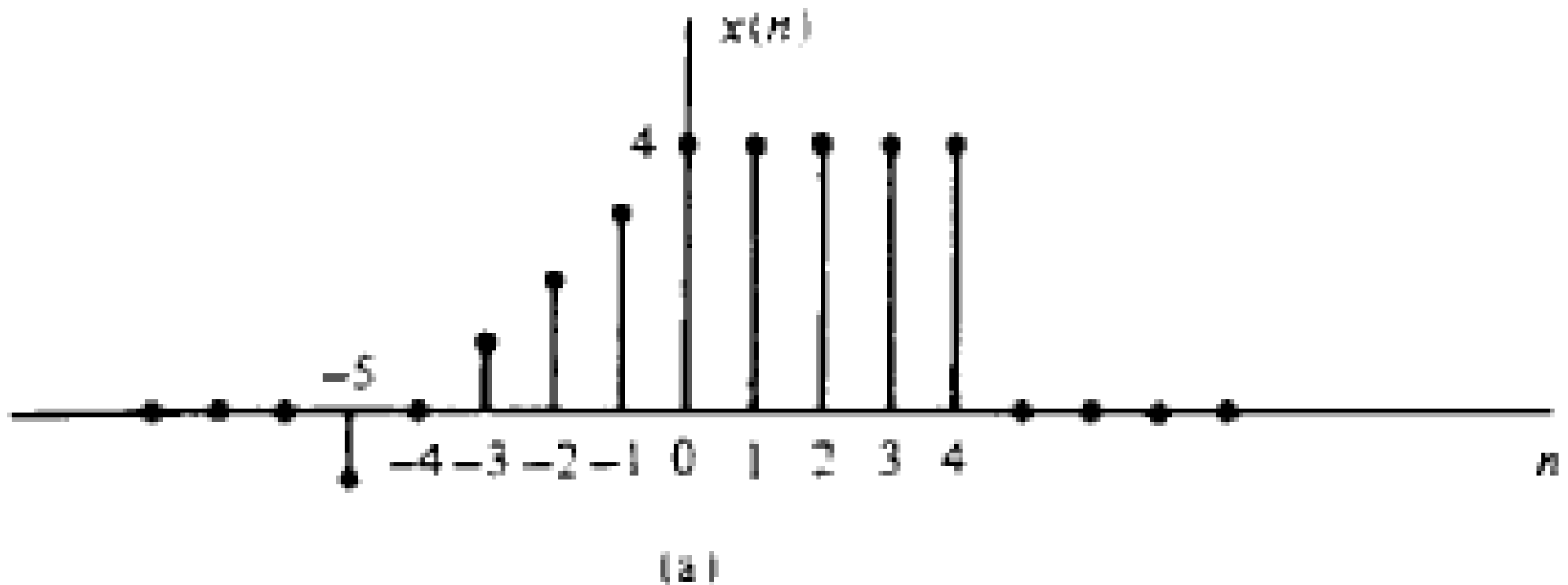
2.1.3 Simple Manipulations of Discrete-Time Signals

In this section we consider some simple modifications or manipulations involving the independent variable and the signal amplitude (dependent variable).

Transformation of the independent variable (time). A signal $x(n)$ may be shifted in time by replacing the independent variable n by $n - k$, where k is an integer. If k is a positive integer, the time shift results in a delay of the signal by k units of time. If k is a negative integer, the time shift results in an advance of the signal by $|k|$ units in time.

Example 2.1.2

A signal $x(n]$ is graphically illustrated in Fig. 2.9a. Show a graphical representation of the signals $x(n - 3)$ and $x(n + 2)$.



Example 2.10.2

$$y(n) = x(n-3)$$

Step 1: $n-3=0 \Rightarrow n=3$

Step 2: Starting point of new transformed signal $y(n)$ will be $-5+3 = -2$

End point of $y(n)$ will be $4+3 = 7$

Step 3: Find $x(n-3)$ now:

$x(n-3)$	$y(n)$
$-2-3 = -5$	-2
-4	-1
-3	0
-2	1
-1	2
0	3
1	4
2	5
3	6
4	7

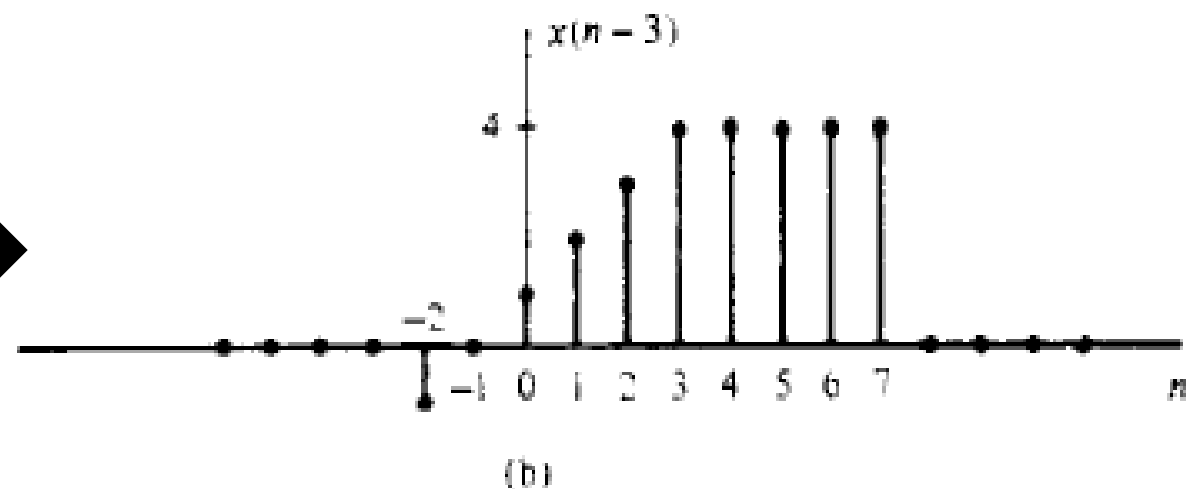
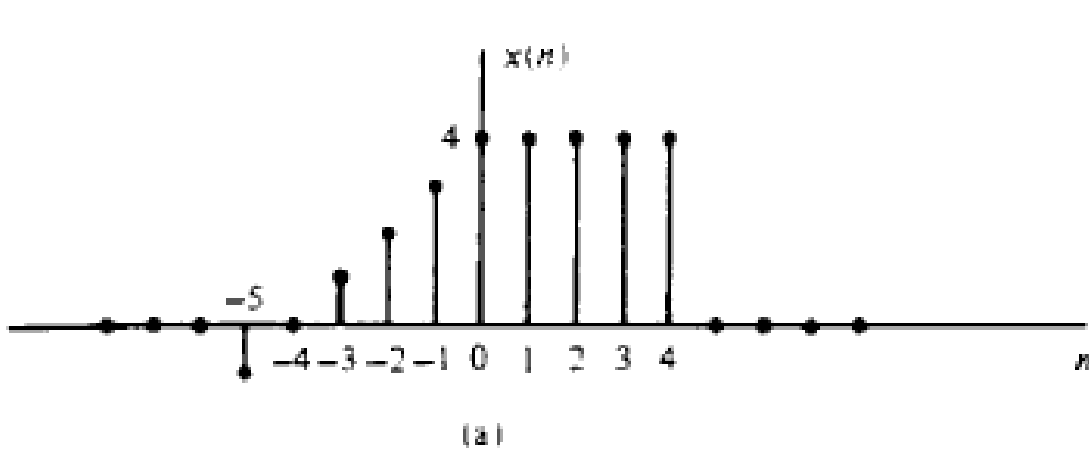
Hence, the "value" at $x(-5)$ will be at $y(-2)$

These are the indices, not amplitudes.

Find $x(n-3]$ now:

$x(n-3)$	$y(n)$
$-2-3=-5$	-2
-4	-1
-3	0
-2	1
-1	2
0	3
1	4
2	5
3	6
4	7

Hence, the "value" at $x(-5)$ will be at $y(-2)$



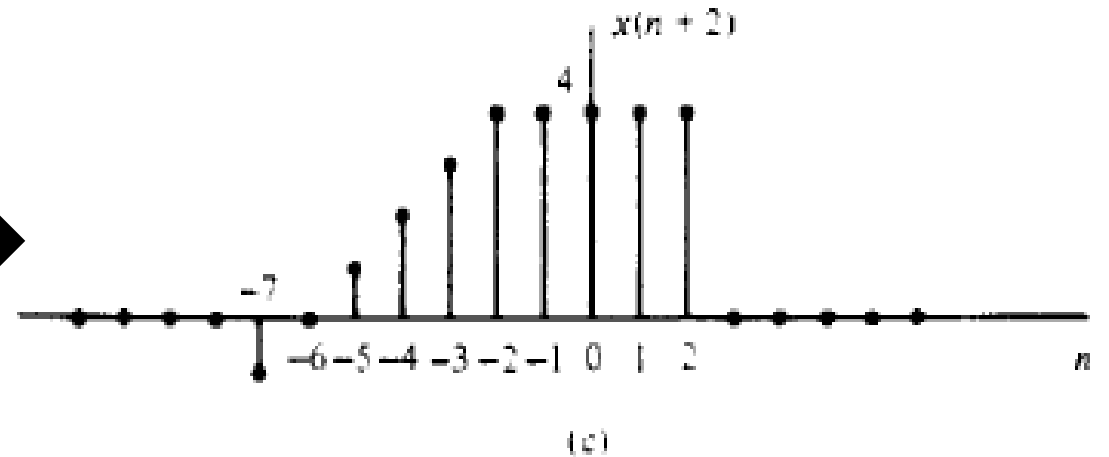
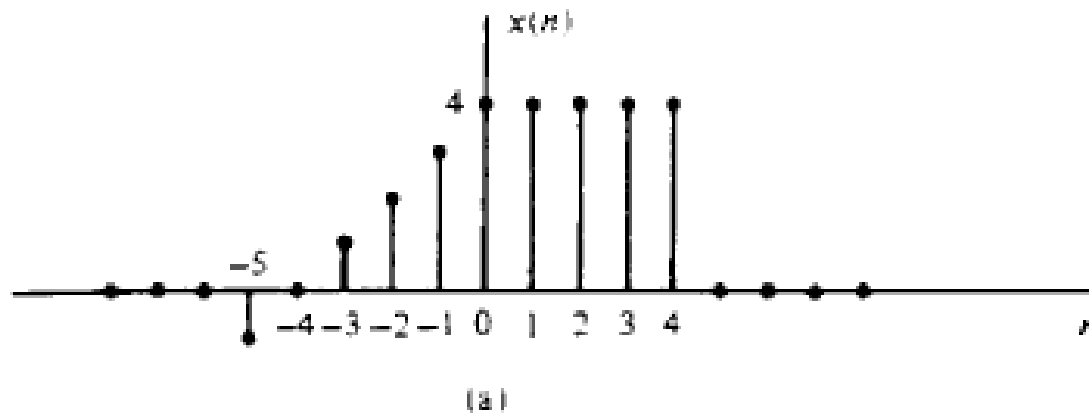
$$y(n) = x(n+2)$$

Step 1: $n+2=0 \Rightarrow n=-2$

Step 2: Range of $y(n)$ will be from -7 to 2.

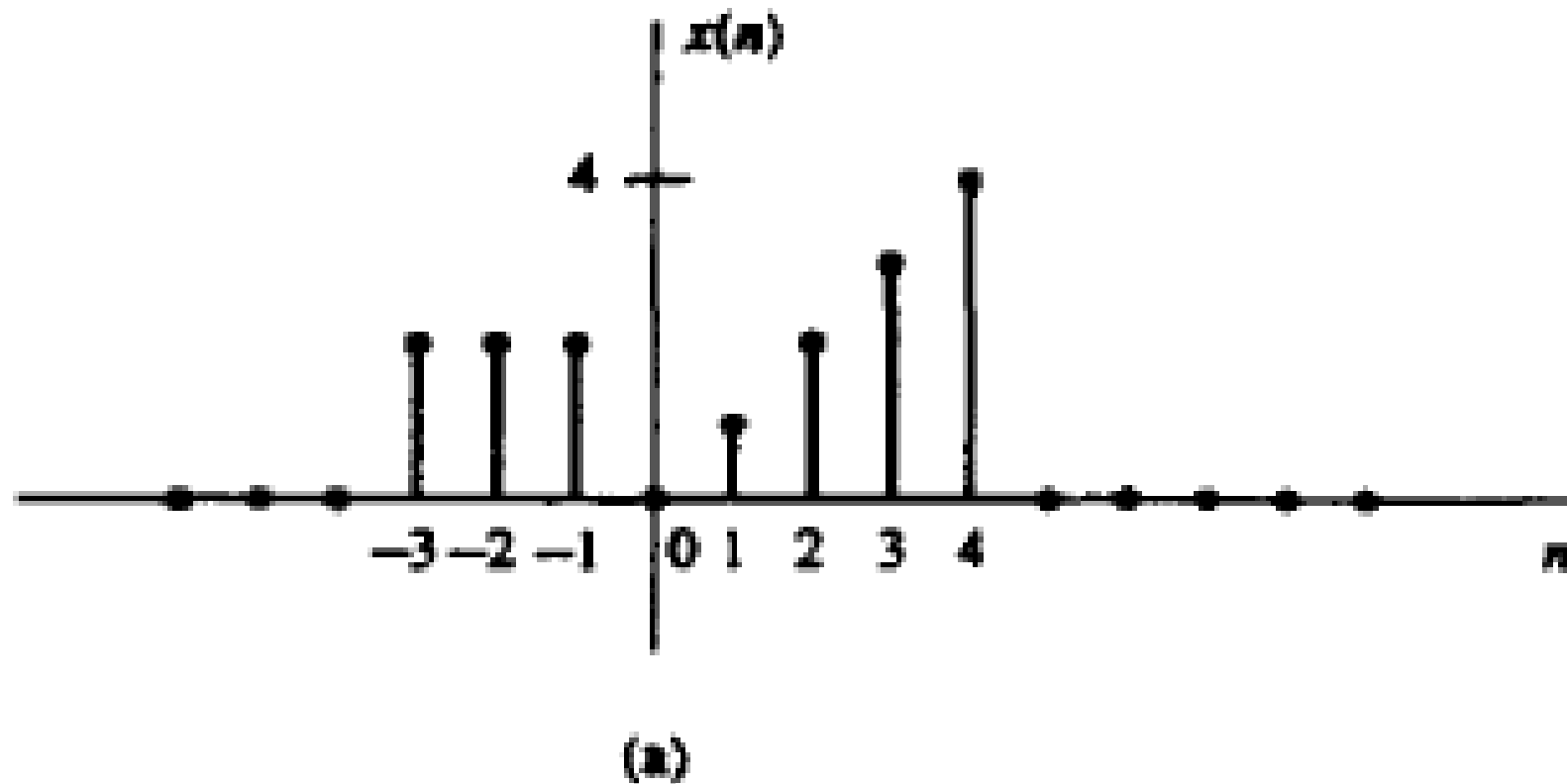
Step 3: Find $x(n+2)$ now:

$x(n+2)$	$y(n)$
$x(-7+2=-5)$	-7
-4	-6
-3	-5
-2	-4
-1	-3
0	-2
1	-1
2	0
3	1
4	2



Example 2.1.3

Show the graphical representation of the signal $x(-n)$ and $x(-n + 2)$, where $x(n)$ is the signal illustrated in Fig. 2.10a.



Solution The new signal $y(n) = x(-n)$ is shown in Fig. 2.10b. Note that $y(0) = x(0)$, $y(1) = x(-1)$, $y(2) = x(-2)$, and so on. Also, $y(-1) = x(1)$, $y(-2) = x(2)$, and so on. Therefore, $y(n)$ is simply $x(n)$ reflected or folded about the time origin $n = 0$. The signal $y(n) = x(-n + 2)$ is simply $x(-n)$ delayed by two units in time. The resulting signal is illustrated in Fig. 2.10c. A simple way to verify that the result in Fig. 2.10c is correct is to compute samples, such as $y(0) = x(2)$, $y(1) = x(1)$, $y(2) = x(0)$, $y(-1) = x(3)$, and so on.

$$y(n) = x(-n+2)$$

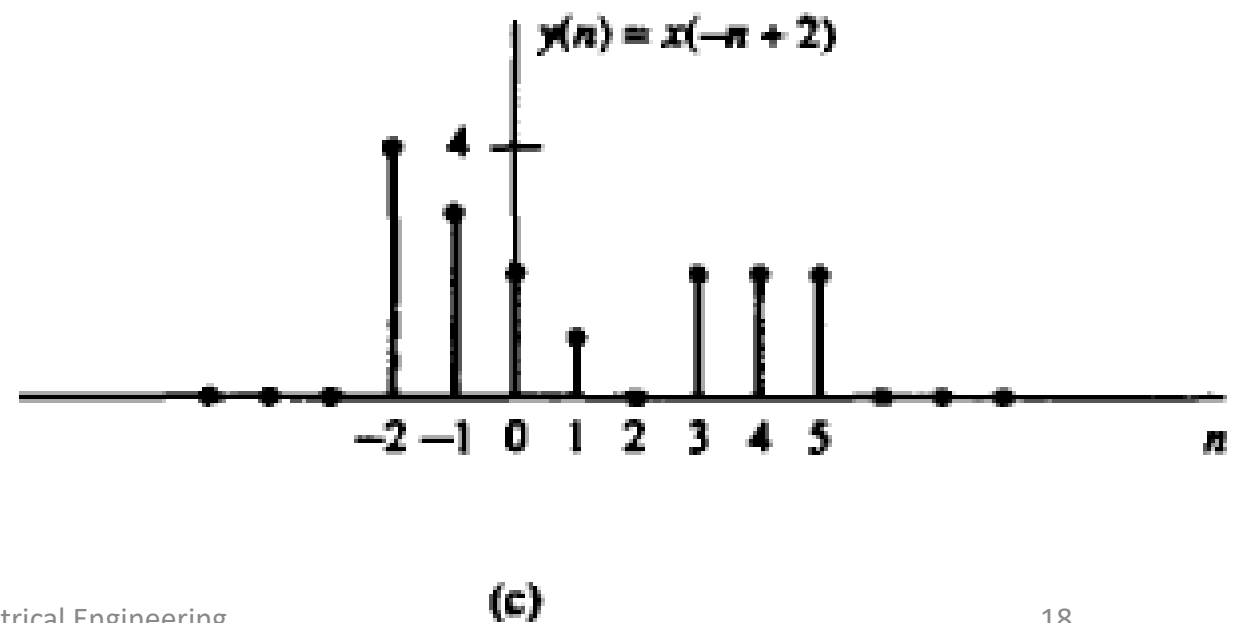
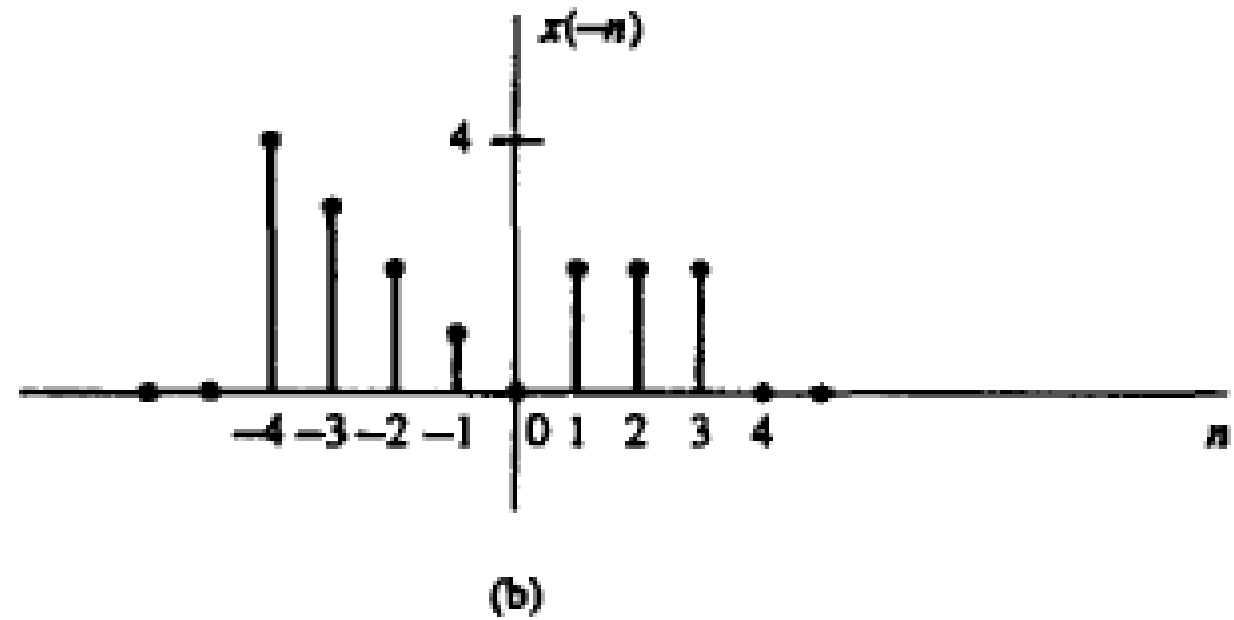
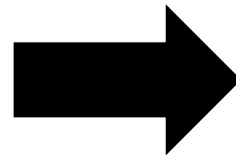
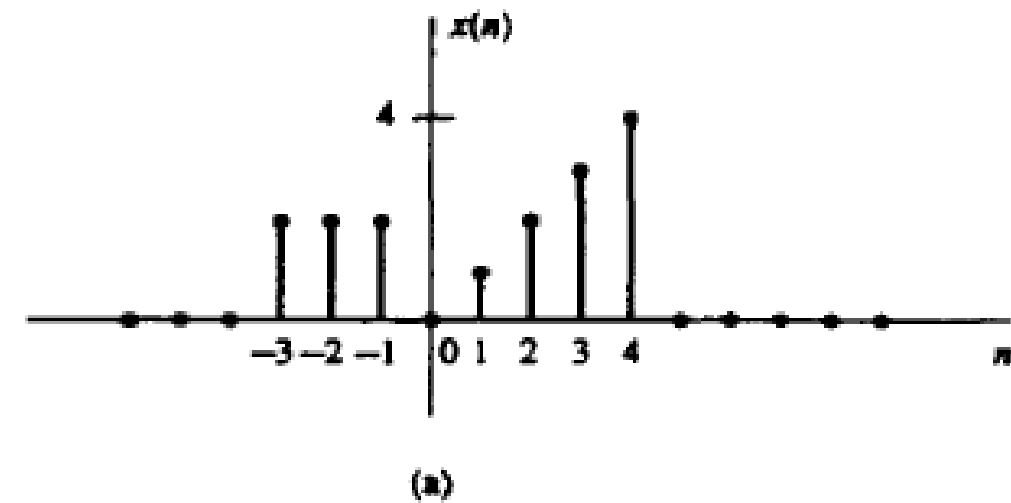
Step 1: $-n+2=0 \Rightarrow n=2$

Step 2: $-2 \leq y(n) \leq 5$

Step 3: Find new transformed signal.

$x(-n+2)$	$y(n)$	
$x(-(-2)+2) = x(4)$	-2	The value at $x(4)$ will be at $y(-2)$.
$x(3)$	-1	" " " " $x(3)$ " " " $y(-1)$.
$x(2)$	0	
$x(1)$	1	
$x(0)$	2	The value at $x(0)$ will be at $y(2)$.
$x(-1)$	3	
$x(-2)$	4	
$x(-3)$	5	

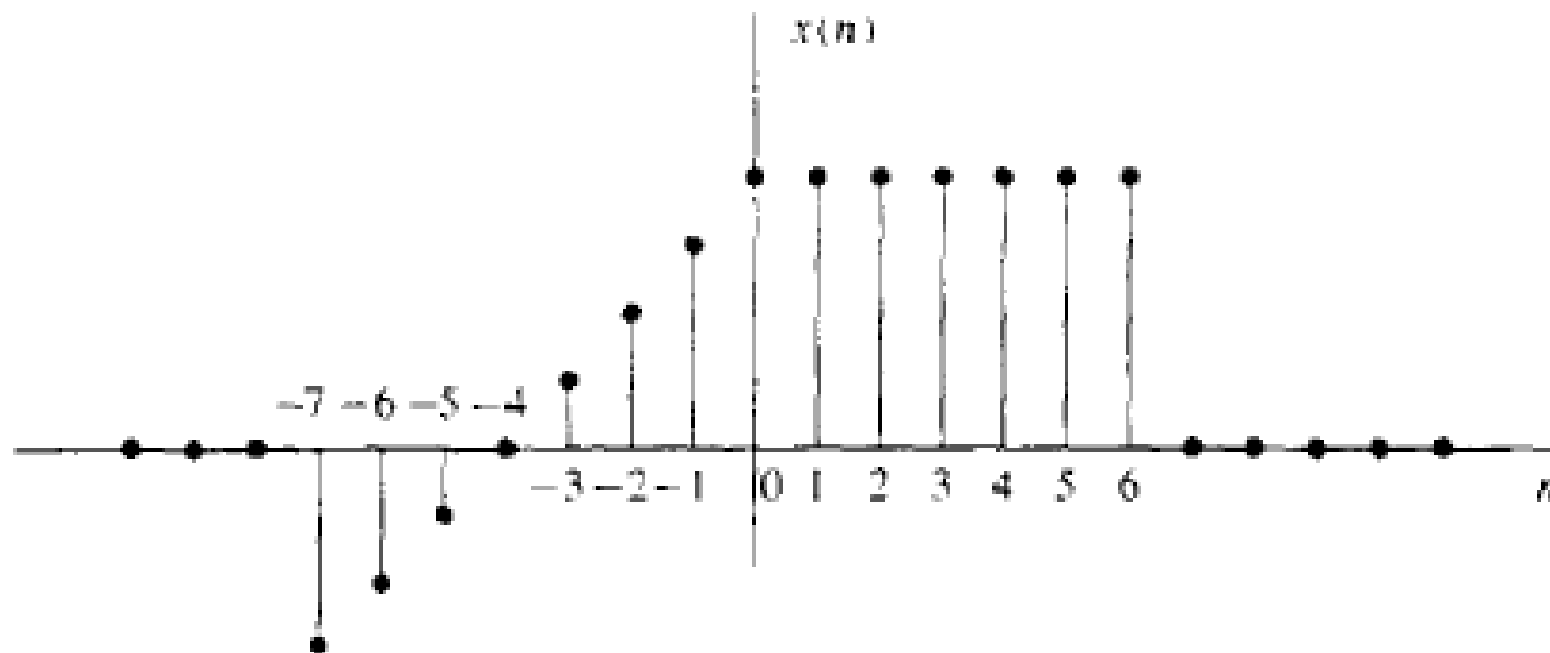




A third modification of the independent variable involves replacing n by μn , where μ is an integer. We refer to this time-base modification as *time scaling* or *down-sampling*.

Example 2.14

Show the graphical representation of the signal $y(n) = x(2n)$, where $x(n)$ is the signal illustrated in Fig. 2.11a.



Take 2 minutes and first try it by yourself.

Solution We note that the signal $y(n)$ is obtained from $x(n)$ by taking every other sample from $x(n)$, starting with $x(0)$. Thus $y(0) = x(0)$, $y(1) = x(2)$, $y(2) = x(4)$, ... and $y(-1) = x(-2)$, $y(-2) = x(-4)$, and so on. In other words, we have skipped the odd-numbered samples in $x(n)$ and retained the even-numbered samples. The resulting signal is illustrated in Fig. 2.11b.

If the signal $x(n)$ was originally obtained by sampling an analog signal $x_a(t)$, then $x(n) = x_a(nT)$, where T is the sampling interval. Now, $y(n) = x(2n) = x_a(2Tn)$. Hence the time-scaling operation described in Example 2.1.4 is equivalent to changing the sampling rate from $1/T$ to $1/2T$, that is, to decreasing the rate by a factor of 2. This is a *downsampling* operation.

Example 2.1.4

$$y(n) = x(2n)$$

Step 1: $n=0$

Step 2: $-7 \leq y(n) \leq 6$

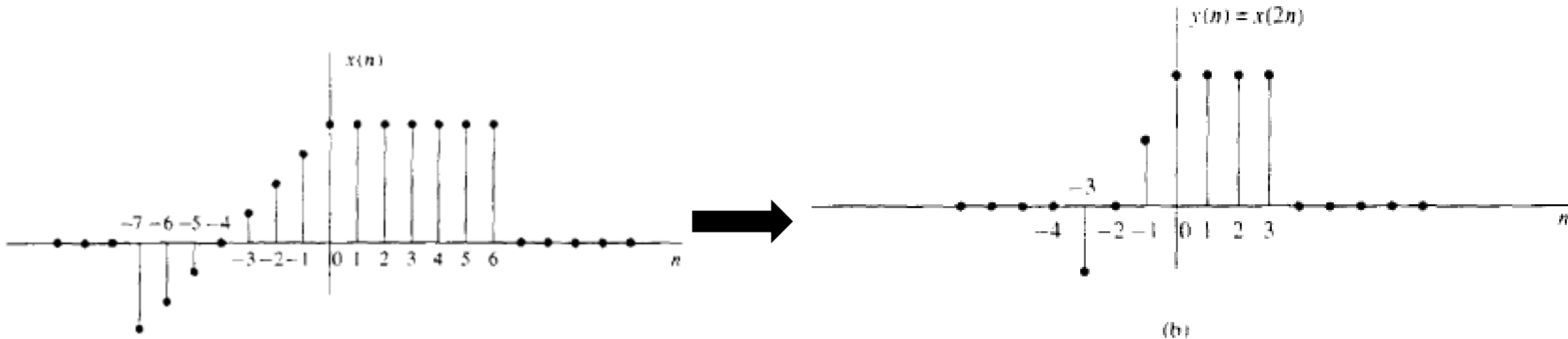
Step 3:

	$x(2n)$	$y(n)$
No values hence discard.	$x(2(-7) = -14)$	-7
	$x(-12)$	-6
	$x(-10)$	-5
	$x(-8)$	-4
	$x(-6)$	-3
	$x(-4)$	-2
	$x(-2)$	-1
	$x(0)$	0
	$x(2)$	1
	$x(4)$	2
No values hence discard.	$x(6)$	3
	$x(8)$	4
	$x(10)$	5
	$x(12)$	6

The value at $x(-6)$ will be at $y(-3)$.
" " " $x(-4)$ " " " $y(-2)$.

	$x(2n)$	$y(n)$
No values hence discard.	$x(2(-7) = -14)$	-7
	$x(-12)$	-6
	$x(-10)$	-5
	$x(-8)$	-4
	$x(-6)$	-3
	$x(-4)$	-2
	$x(-2)$	-1
	$x(0)$	0
	$x(2)$	1
	$x(4)$	2
	$x(6)$	3
No values hence discard.	$x(8)$	4
	$x(10)$	5
	$x(12)$	6

The value at $x(-6)$ will be at $y(-3)$.
 " " " $x(-4)$ " " " $y(-2)$.



Addition, multiplication, and scaling of sequences. Amplitude modifications include *addition*, *multiplication*, and *scaling* of discrete-time signals.

Amplitude scaling of a signal by a constant A is accomplished by multiplying the value of every signal sample by A . Consequently, we obtain

$$y(n) = Ax(n) \quad -\infty < n < \infty$$

The *sum* of two signals $x_1(n)$ and $x_2(n)$ is a signal $y(n)$, whose value at any instant is equal to the sum of the values of these two signals at that instant, that is,

$$y(n) = x_1(n) + x_2(n) \quad -\infty < n < \infty$$

The *product* of two signals is similarly defined on a sample-to-sample basis as

$$y(n) = x_1(n)x_2(n) \quad -\infty < n < \infty$$

2.2 DISCRETE-TIME SYSTEMS

In many applications of digital signal processing we wish to design a device or an algorithm that performs some prescribed operation on a discrete-time signal.

Such a device or algorithm is called a discrete-time system. More specifically, a *discrete-time system* is a device or algorithm that operates on a discrete-time signal, called the *input* or *excitation*, according to some well-defined rule, to produce another discrete-time signal called the *output* or *response* of the system.

We say that the input signal $x(n)$ is *transformed* by the system into a signal $y(n)$, and express the general relationship between $x(n)$ and $y(n)$ as

$$y(n) \equiv \mathcal{T}[x(n)] \quad (2.2.1)$$

where the symbol $\mathcal{T}$ denotes the transformation (also called an operator), or processing performed by the system on $x(n)$ to produce $y(n)$. The mathematical relationship in (2.2.1) is depicted graphically in Fig. 2.12.

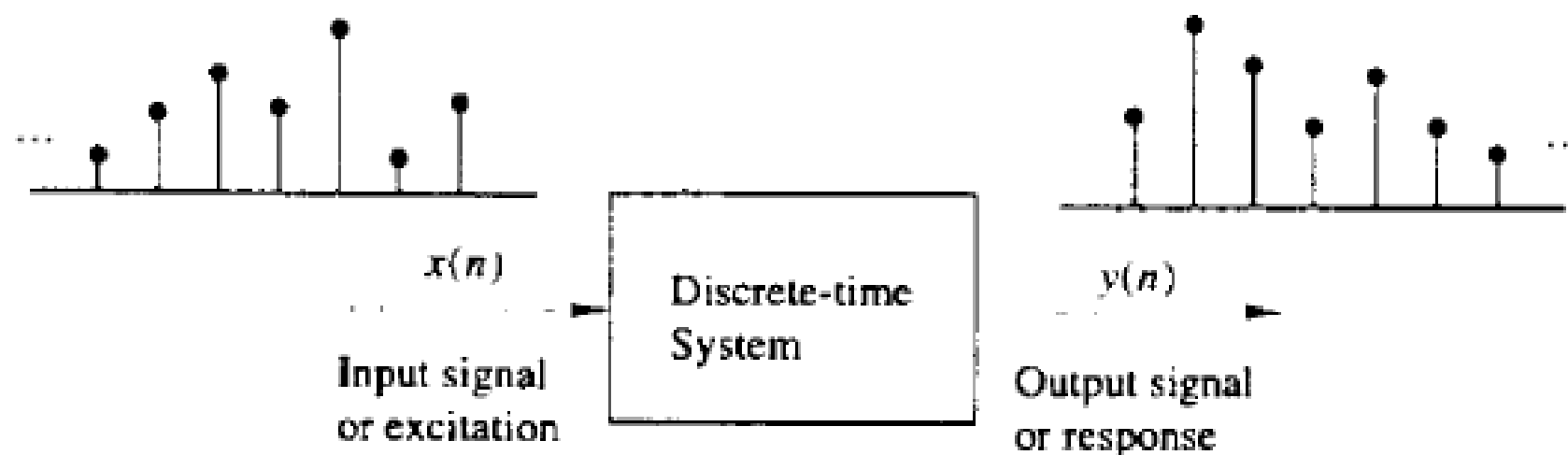


Figure 2.12 Block diagram representation of a discrete-time system.

2.2.1 Input–Output Description of Systems

The input–output description of a discrete-time system consists of a mathematical expression or a rule, which explicitly defines the relation between the input and output signals (*input–output relationship*). The exact internal structure of the system is either unknown or ignored. Thus the only way to interact with the system is by using its input and output terminals (i.e., the system is assumed to be a “black box” to the user).

the general input-output relationship in (2.2.1) or, alternatively, the notation

$$x(n) \xrightarrow{\mathcal{T}} y(n) \quad (2.2.2)$$

which simply means that $y(n)$ is the response of the system $\mathcal{T}$ to the excitation $x(n)$.

Example 2.2.1

Determine the response of the following systems to the input signal

$$x(n) = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

(a) $y(n) = x(n)$

- (a) In this case the output is exactly the same as the input signal. Such a system is known as the *identity* system.

$$\textbf{(b)} \quad y(n) = x(n - 1)$$

(b) This system simply delays the input by one sample. Thus its output is given by

$$x(n) = \{ \dots, 0, 3, 2, 1, 0, 1, 2, 3, 0, \dots \}$$

↑

$$\textbf{(c)} \quad y(n) = x(n + 1)$$

(c) In this case the system “advances” the input one sample into the future. For example, the value of the output at time $n = 0$ is $y(0) = x(1)$. The response of this system to the given input is

$$x(n) = \{ \dots, 0, 3, 2, 1, 0, 1, 2, 3, 0, \dots \}$$

↑

$$(d) \quad y(n) = \frac{1}{3}[x(n+1) + x(n) + x(n-1)]$$

(d) The output of this system at any time is the mean value of the present, the immediate past, and the immediate future samples. For example, the output at time $n = 0$ is

$$y(0) = \frac{1}{3}[x(-1) + x(0) + x(1)] = \frac{1}{3}[1 + 0 + 1] = \frac{2}{3}$$

Repeating this computation for every value of n , we obtain the output signal

$$y(n) = \{ \dots, 0, 1, \frac{5}{3}, 2, 1, \frac{2}{3}, 1, 2, \frac{5}{3}, 1, 0, \dots \}$$

↑

$$(e) \quad y(n) = \max\{x(n+1), x(n), x(n-1)\}$$

$$(f) \quad y(n) = \sum_{k=-\infty}^n x(k) = x(n) + x(n-1) + x(n-2) + \dots$$

(e) This system selects as its output at time n the maximum value of the three input samples $x(n-1)$, $x(n)$, and $x(n+1)$. Thus the response of this system to the input signal $x(n)$ is

$$y(n) = \{0, 3, 3, 3, 2, 1, 2, 3, 3, 3, 0, \dots\}$$

↑

(f) This system is basically an *accumulator* that computes the running sum of all the past input values up to present time. The response of this system to the given input is

$$y(n) = \{\dots, 0, 3, 5, 6, 6, 7, 9, 12, 0, \dots\}$$

↑

Consider, for instance, the accumulator in the example. We see that the output at time $n = n_0$ depends not only on the input at time $n = n_0$, but also on $x(n)$ at times $n = n_0 - 1, n_0 - 2$, and so on. By a simple algebraic manipulation the input-output relation of the accumulator can be written as

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^n x(k) = \sum_{k=-\infty}^{n-1} x(k) + x(n) \\ &= y(n-1) + x(n) \end{aligned} \tag{2.2.4}$$

which justifies the term *accumulator*. Indeed, the system computes the current value of the output by adding (accumulating) the current value of the input to the previous output value.

Example 2.2.2

The accumulator described by (2.2.3) is excited by the sequence $x(n) = nu(n)$. Determine its output under the condition that:

- (a)** It is initially relaxed [i.e., $y(-1) = 0$].
- (b)** Initially, $y(-1) = 1$.

Solution The output of the system is defined as

$$\begin{aligned}y(n) &= \sum_{k=-\infty}^n x(k) = \sum_{k=-\infty}^{-1} x(k) + \sum_{k=0}^n x(k) \\&= y(-1) + \sum_{k=0}^n x(k)\end{aligned}$$

But

$$\sum_{k=0}^n x(k) = \frac{n(n+1)}{2}$$

(a) If the system is initially relaxed, $y(-1) = 0$ and hence

$$y(n) = \frac{n(n+1)}{2} \quad n \geq 0$$

(b) On the other hand, if the initial condition is $y(-1) = 1$, then

$$y(n) = 1 + \frac{n(n+1)}{2} = \frac{n^2 + n + 2}{2} \quad n \geq 0$$